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1. Introduction — The Need for FANS

> **What this section covers:** Why traditional voice-based ATC has limitations over oceanic/remote areas, and why a satellite-based integrated system became necessary.

Aircraft are currently controlled using voice communications. Over and close to populated landmasses, ATC uses radar with VHF communications. However, over **oceans, deserts and polar regions**, VHF and radar may not be available. ATC must provide a **procedural control service** using HF communications, resulting in high vertical, lateral and longitudinal separation distances and low traffic density.

Position reports are passed by aircraft crossing the North Atlantic every **10° of longitude** up to **70°N** and every **20°** north thereof — meaning ATC receive a position update every **30–60 minutes**.

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> **Core Problem:** Pilot rarely communicates directly with ATC controller over oceans — messages relayed via a third party using HF. Static interference, fading, and message repetition compound the problem. Large separation is required because position updates are infrequent and unreliable.

Advances in technology now provide:

- **GNSS** — global satellite navigation providing position accuracy better than **1 NM**
- **SATCOM** — satellite communications providing potential for global communications through a single medium

FANS integrates these into a Seamless Global Air Traffic Management System.

> **Exam Tip:** The 30–60 minute position update interval and the use of HF with a relay third party are key exam facts that illustrate why FANS was developed.

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2. Communications Systems

> **What this section covers:** The evolution from VHF/HF voice to SATCOM, and the disadvantages of voice communications.

Until the late 1980s, air-ground communications relied solely on VHF and HF voice. In the 1990s, voice was extended to SATCOM (UHF) via geostationary satellites. Coverage is limited to approximately **80° of latitude**. Polar regions require satellites in lower-altitude inclined orbits.

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Medium	Range	Quality	Key Limitation
VHF	Line of sight (~**200 NM** at FL300)	Good	Not available over oceans
HF	Long range (sky waves)	Poor	Static, fading, third-party relay, SELCAL needed
SATCOM (UHF)	Global to ~**80° latitude**	Good	No polar coverage; geostationary only

Disadvantages of Voice Communications (DGCA list):

- Many aircraft on one frequency (congestion)
- Synchronous transmission (only one transmits at a time)
- Language confusion
- Limited channels
- VHF — line of sight only
- HF — interference, tiring to listen to

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3. Data Link

What this section covers: Data link definition, providers, airborne equipment, and UL/DL terminology.

A **data link** is a means of connecting one location to another for transmitting and receiving information. Data links may be established on any frequency but require additional equipment on both ground and aircraft.

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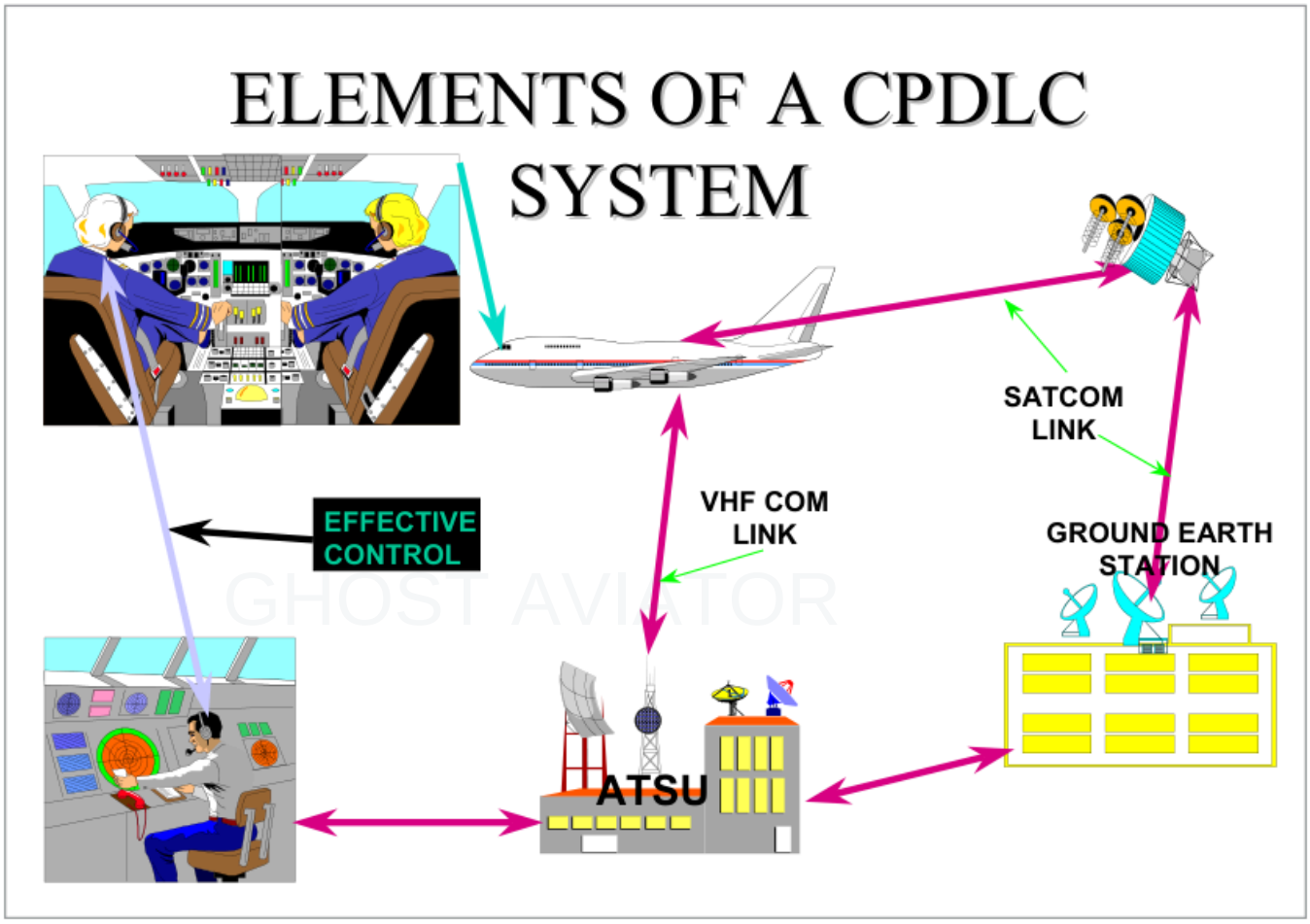


Fig 24.2 — FANS A CPDLC system elements

Two Data Link Service Providers:

- **SITA** — Société Internationale de Télécommunications Aéronautiques (France) — operates ATN (FANS B)
- **ARINC** — Air Radio Incorporated (USA) — operates ACARS network (FANS A)

Uplink / Downlink Convention

Direction	Definition
Uplink (UL)	Transmission from earth station/ATSU to aircraft
Downlink (DL)	Transmission from aircraft to earth station/ATSU (even if aircraft is on the ground)

Memory Aid: "UP from the ground, DOWN from the aircraft." The aircraft's message is always the Downlink regardless of whether it is airborne or on the ground.

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Airborne Data Link Equipment

Unit	Full Name	Function
CMU	Communications Management Unit	Selects frequencies for all radio equipment
DCDU	Data Communications Display Unit	Displays messages received/sent via data link
MCDU	Multi Control and Display Unit	Combination of CMU and DCDU
VISUAL	Attention Getters	Light & sound alert for incoming data link message
PRINTER	Hard Copy Output	Prints data link messages in cockpit

4. ACARS

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> **What this section covers:** ACARS — the first major aviation data link system, its evolution, and capabilities.

ACARS (Aircraft Communications Addressing and Reporting System) uses data link format to pass messages between the aircraft and ATC or aircraft operating companies using VHF. Messages can be printed out in the cockpit.

- Extended to **SATCOM** in the early **1990s** for oceanic flights
- Extended to **HF** in **2001** to close the polar gap
- FMS interfaces added in the 1990s — met info, alternate routes evaluated by FMS
- Maintenance monitoring via ATSU/DCDU

> **ACARS Coverage Evolution:** VHF (populated areas) → + SATCOM (1990s, global except poles) → + HF (2001, polar) = Near-global coverage.

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5. Air Traffic Service Unit (ATSU) & CNS/ATM

> **What this section covers:** ATSU services and the CNS/ATM concept.

The **ATSU (Air Traffic Service Unit)** provides:

- Flight Information Service
- Alerting Service
- Air Traffic Advisory Service
- Air Traffic Control Service (Area / Approach / Aerodrome)

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CNS/ATM uses automation, digital technology and satellite systems to give a Seamless Global Air Traffic Management System.

Acronym	Facilities	Operated By
CNS	Communications, Navigation, Surveillance	ATSU (Air Traffic Service Unit)
ATM	Air Traffic Management	AOC (Aircraft Operational Centre = airline)

6. Data Link Message Types

> **What this section covers:** ATM (ATC) and AOC (airline operational) data link message categories with examples.

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ATM (ATC) Data Link Messages

Message Type	Purpose
PDC (Pre-Departure Clearance)	Full departure clearance uplinked before push-back
Oceanic Clearance	Route clearance for oceanic track system
D-ATIS	Automatic Terminal Information Service via data link
CPDLC	Controller–Pilot Data Link Communications — in-flight ATC instructions
Free Text	Unformatted messages (e.g., re-routing)

AOC (Airline) Data Link Messages — OOOI

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Code	Event	Example Message
OUT	Out of gate	OUTRP 1065/31 EGLL/KJFK .G-BOAC/OUT 0850
OFF	Off ground (airborne)	OFFRP .../OFF 0900/ETA 1525
ON	On ground (landed)	ONRP .../ON 1540
IN	In the gate	INRP .../IN 1550

Additional AOC messages: Load Sheet, Passenger Information, Weather Reports (METAR/TAF/SIGMET/VOLMET), Maintenance Reports.

Exam Tip — OOOI: "Out, Off, On, In" — the four flight-event reports automatically sent to the airline so they know where each aircraft is throughout its cycle.

7. Logon Procedure

What this section covers: How an aircraft connects to the CPDLC/FANS data link service via the ATS Facilities Notification (AFN).

```
flowchart LR
  A["Pilot inputs 4-digit ICAO\nATSU address into FMS"] --> B["FMS sends Logon Message\n(aircraft address + capabilities)"]
  B --> C["ATSU acknowledges\nlogon message"]
  C --> D["ATSU sends Connection\nRequest to aircraft"]
  D --> E["Aircraft sends Connection\nConfirm message"]
  E --> F["AFN Complete\n(ATS Facilities Notification)"]
```

Manual Logon Required When:

- First contact with the ATSU on the ground
- Entering a CPDLC area from a non-CPDLC area
- Any interruption to the service (link broken)

Once established, automatic transfer to subsequent CPDLC-capable ATSUs.

Direction Convention — Critical: Aircraft → ATSU = *Downlink (DL)*. ATSU → Aircraft = *Uplink (UL)*. This applies even when the aircraft is on the ground.

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8. FANS A — Oceanic / Remote Airspace

What this section covers: FANS A components — AFN, ADS-C, and CPDLC.

FANS A provides a CNS system and ADS. Used over **oceanic and remote airspace**, transmitted over the **ACARS network (ARINC)**. Communications use current HF/VHF frequencies; GNSS provides the navigation input for surveillance.

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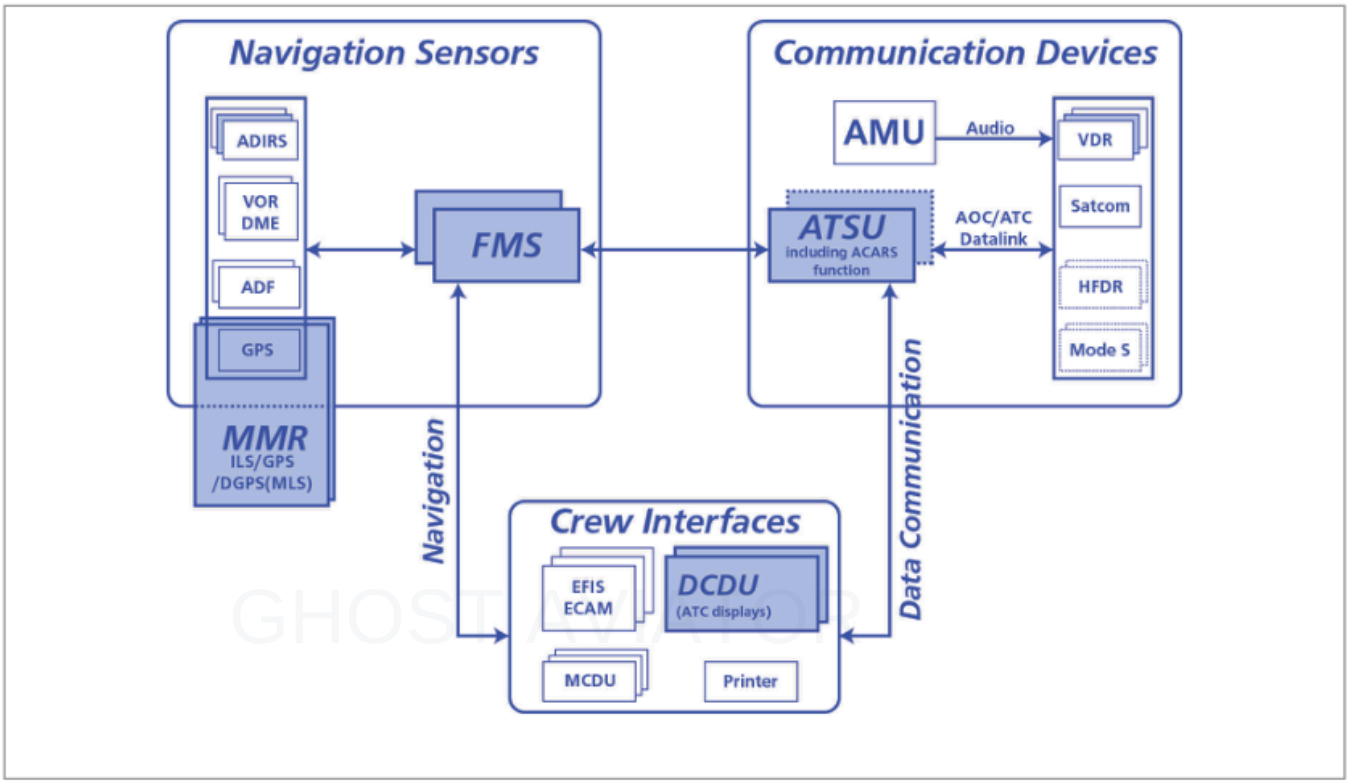


Fig 24.4 — Typical FANS A architecture

FANS A Components

AFN — ATS Facility Notification

A contact message initiated by aircrew or automatic aircraft trigger. If acknowledgement is not received within a pre-set time, or there is an erroneous reply, an error message is displayed to the aircrew.

ADS-C — Automatic Dependent Surveillance – Contract

Controller-set contract with the aircraft's FMS, without any pilot input. The flight crew have **no workload** associated with setup.

Contract Type	Description
Periodic	Position reports at regular time intervals
On Demand	ATSU requests a single position report
On Event	Aircraft reports when specific event occurs (e.g., waypoint passage)
Emergency Mode	High-rate reporting during declared emergency

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Critical Rule: Only the *flight crew* can declare and cancel ADS-C emergency reporting. The aircraft cannot initiate a contract itself.

CPDLC — Controller Pilot Data Link Communications

CPDLC permits data link messages for all stages of flight. Fixed-format messages activated by ATC controller or pilot. Messages annotated whether a response is required. An unanswered message (e.g. "report levelling at FL310") remains open until the FMS sends the automatic response.



Fig 24.5 — Typical DCDU message display

Valid ATSU Combinations (FANS A):

- AFN Only, or ADS Only, or CPDLC Only
- AFN and CPDLC
- ADS and CPDLC

NOT permitted: AFN and ADS together.

9. FANS B — High Density Airspace

What this section covers: FANS B differences from FANS A, and the ATN.

FANS B is very similar to FANS A but operates within **high density airspace** with good VHF coverage. Operates over the **Aeronautical Telecommunications Network (ATN)**, operated by **SITA**. The ATN allows ground/ground, ground/air and avionic data subnetworks to interoperate.

Feature	FANS A	FANS B
Airspace	Oceanic / Remote	High Density (continental)
Network Operator	ARINC	SITA
Network Used	ACARS	ATN
Primary Comms	HF, VHF, SATCOM	VHF (good coverage area)

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Memory Aid: FANS **A** = **A**RINC = **A**ceanic. FANS **B** = **B**usy/high density = SITA/ATN.

10. Useful Abbreviations

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Abbreviation	Full Form
ACARS	Aircraft Communications Addressing and Reporting System
ADS / ADS-C	Automatic Dependent Surveillance / Contract
AFN	Air Traffic Facilities Notification
AOC	Airline Operational Centre
ARINC	Air Radio Incorporated (USA)
ATM	Air Traffic Management
ATN	Aeronautical Telecommunications Network
ATSU	Air Traffic Service Unit
CNS	Communication, Navigation and Surveillance
CPDLC	Controller Pilot Data Link Communications
D-ATIS	Data Link Air Terminal Information Service
DCDU	Data Link Control and Display Unit
DCL	Departure Clearance
FANS A	Data Link Package for Oceanic/Remote airspace (ARINC/ACARS)
FANS B	Data Link Package for High Density airspace (SITA/ATN)
MCDU	Multi-function Control and Display Unit
OCL	Oceanic Clearance
OOOI	Out of gate, Off the ground, On the ground, In the gate
SITA	Société Internationale de Télécommunications Aéronautiques (France)
VDL2	VHF Data Link Mode 2

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Quick Revision Summary — Chapter 24:

- FANS solves oceanic ATC limitations: no radar/VHF → procedural control, 30–60 min updates, HF relay
- Data link overcomes voice disadvantages: digital, non-synchronous, printable
- FANS A (ARINC/ACARS) = oceanic; FANS B (SITA/ATN) = high-density continental
- OOOI = Out, Off, On, In — AOC flight event messages
- Downlink = aircraft to ATSU; Uplink = ATSU to aircraft (even on ground)
- ADS-C: controller sets contract, crew only declare/cancel emergency mode
- SATCOM coverage limited to ±80° latitude; HF added to ACARS for polar in 2001
- AFN = logon process: manual first time, then automatic handoff to next ATSU

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Practice Questions & Detailed Answers

Instructor-generated questions in DGCA CPL/ATPL examination style.

Q1. What is the primary limitation of ATC over oceanic regions that FANS is designed to overcome?

a. Excessive radar coverage requiring too many controllers

b. Lack of VHF/radar leading to procedural separation and infrequent position updates

c. Overloading of SATCOM frequency bands

d. Pilot inability to communicate in English

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Correct Answer: **(b)**

Explanation: Over oceans, deserts and polar regions, VHF (line-of-sight) and radar are unavailable. ATC must provide procedural service using HF, resulting in large separations and position reports only every 30–60 minutes. FANS solves this with GNSS position accuracy and satellite data links for real-time communication. See Section 1.

Why other options are wrong:

- **(a)** Oceanic areas have too few controllers due to communication difficulty — the opposite of this option.
- **(c)** SATCOM band congestion is not the described problem.
- **(d)** Language confusion is a minor voice comms disadvantage, not the primary oceanic limitation.

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Instructor's Note: *The 30–60 minute update interval is the key figure that underscores the urgency of FANS development.*

Q2. FANS A operates over which network and is it operated by whom?

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a. ATN, operated by SITA

b. ACARS network, operated by ARINC

c. SATCOM network, operated by INMARSAT

d. VHF Data Link Mode 2, operated by Eurocontrol

Correct Answer: **(b)**

Explanation: FANS A is designed for oceanic and remote airspace and is transmitted over the ACARS network operated by ARINC (USA). FANS B uses the ATN operated by SITA. See Section 8 and Section 9.

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Why other options are wrong:

- **(a)** ATN/SITA is FANS B (high density), not FANS A.
- **(c)** INMARSAT is a SATCOM provider but not the network name for FANS A.
- **(d)** VDL2 is a data link medium, not the FANS A network.

Instructor's Note: *FANS A = ARINC/ACARS/Oceanic. FANS B = SITA/ATN/Continental. This distinction is a classic exam question.*

Q3. What does OOOI stand for?

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a. On Approach, On Descent, On Instruments, In Cloud

b. Out of gate, Off the ground, On the ground, In the gate

c. Over ocean, Over obstacle, On ILS, Inside marker

d. On track, Off track, On altitude, Inside airspace

Correct Answer: **(b)**

Explanation: OOOI messages are the four key flight event reports sent automatically to the AOC via

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data link, allowing the airline to track each aircraft through its full cycle. See Section 6.

Why other options are wrong:

- **(a), (c), (d)** — Fictitious mnemonics not associated with ACARS/AOC terminology.

Instructor's Note: *OOOI reports are also used by maintenance to track engine cycles and aircraft utilization.*

Q4. Regarding ADS-C, which statement is correct?

a. The aircraft automatically initiates all ADS-C contracts without crew input

b. The controller sets up the contract; flight crew have no associated workload

c. The pilot must manually update ADS-C position every 10 minutes

d. ADS-C emergency mode can be declared by either the ATSU or the flight crew

Correct Answer: **(b)**

Explanation: ADS-C is a controller-set contract with the aircraft's FMS. The controller establishes all contract parameters without any pilot input — flight crew have zero workload. See Section 8.

Why other options are wrong:

- **(a)** The aircraft cannot initiate a contract; it only responds to controller-initiated ones.
- **(c)** Manual updates are not required; the system is fully automatic.
- **(d)** Only flight crew can declare/cancel ADS-C emergency reporting — not the ATSU.

Instructor's Note: *The exclusive flight-crew control of ADS-C emergency mode declaration is a specific DGCA exam point.*

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Q5. A message transmitted from an aircraft to the ATSU while the aircraft is still on the ground is called a:

a. Uplink, because the aircraft is below the satellite

b. Downlink, because the convention is aircraft-to-ATSU regardless of aircraft position

c. Ground link, a special category for ground operations

d. Uplink, because messages from aircraft always go upward

Correct Answer: **(b)**

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Explanation: In FANS/CPDLC, aircraft-to-ATSU messages are always called Downlink (DL) and ATSU-to-aircraft messages are always Uplink (UL). This is a fixed convention that applies even when the aircraft is on the ground. See Section 7.

Why other options are wrong:

- **(a)** The satellite position is irrelevant to the DL/UL naming convention in CPDLC.
- **(c)** There is no "ground link" category in FANS terminology.
- **(d)** "Upward from aircraft" is a physical description, not the convention used — this confuses physical direction with the CPDLC naming rule.

Instructor's Note: *The "even if on the ground" qualifier is the classic trick in this type of question.*

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Q6. At what approximate latitude does geostationary SATCOM coverage become unreliable?

a. Above FL350 at any latitude

b. Beyond approximately 80° of latitude (polar regions)

c. When aircraft exceeds Mach 0.85

d. Below 10,000 ft altitude

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Correct Answer: **(b)**

Explanation: Geostationary satellites orbit above the equator. Their coverage is limited to approximately **80° of latitude**. Polar regions (beyond 80°) require satellites in lower-altitude inclined orbits to provide coverage. See Section 2.

Why other options are wrong:

- **(a)** Altitude/FL has no impact on SATCOM latitude coverage limits.
- **(c)** Mach number is completely irrelevant to SATCOM satellite geometry.
- **(d)** Low altitude does not affect the latitude coverage of geostationary satellites.

Instructor's Note: *This is why HF was added to ACARS in 2001 and why lower-orbit satellite constellations (LEO) are being developed for polar aviation.*

Master Reference Tables

Key Numerical Values — Chapter 24

Value	Parameter	Section Ref
<1 NM	GNSS position accuracy	1
30–60 min	Position update interval over N. Atlantic (procedural)	1
10° longitude	N. Atlantic position reporting interval (up to 70°N)	1
20° longitude	N. Atlantic position reporting interval (north of 70°N)	1
~200 NM	VHF range from transmitter at FL300	2
~80° latitude	Geostationary SATCOM coverage limit	2
2001	Year HF added to ACARS for polar coverage	4

Q&A Answer Key

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Q	Answer	Key Topic
1	b	Oceanic ATC limitations
2	b	FANS A = ARINC/ACARS
3	b	OOOI messages
4	b	ADS-C crew/controller roles
5	b	Downlink convention
6	b	SATCOM polar coverage limit

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